Week 4: Linear and Nonlinear Models

CCS4004 - Optimization Techniques

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Today's Journey

1. Linear Programming (LP)
2. Nonlinear Programming (NLP)
3. LP vs NLP Comparison
4. Real-World Applications
5. Let's Practice!
6. Software for LP and NLP
7. Summary and Takeaways

Quick Question

Is the real world linear?

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- Doubling ingredients doubles taste quality? Rarely!
- Energy consumption linear with speed? No! (quadratic or cubic)

Then why study linear models?

- Simple, fast, well-understood
- Good approximations in many cases
- Building blocks for nonlinear methods

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What Makes a Model Linear?

Linear Programming (LP)

An optimization problem where both the objective function and all constraints are **linear**.

General Form:

$$
\begin{aligned}
\max \text{ or } \min \quad & c_1x_1 + c_2x_2 + \cdots + c_nx_n \\
\text{subject to} \quad & a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \leq b_1 \\
& a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \leq b_2 \\
& \vdots \\
& x_1, x_2, \ldots, x_n \geq 0
\end{aligned}
$$

Key: No products like x_1x_2, no powers like x_1^2, no $\sin(x)$, e^x, etc.

Linear vs Nonlinear: Can You Tell?

Which are linear?

1. $3x_1 + 5x_2 \leq 100$

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2. $x_1^2 + x_2^2 \leq 25$ ✗ Nonlinear (quadratic)

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1. $3x_1 + 5x_2 \leq 100$ ✓Linear
2. $x_1^2 + x_2^2 \leq 25$ ✗ Nonlinear (quadratic)
3. $2x_1 - 7x_2 + 3x_3 = 15$

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4. $x_1 \cdot x_2 \geq 10$ ✗ Nonlinear (bilinear)
5. $\sqrt{x_1} + x_2 \leq 8$

Linear vs Nonlinear: Can You Tell?

Which are linear?

1. $3x_1 + 5x_2 \leq 100$ ✓Linear
2. $x_1^2 + x_2^2 \leq 25$ **X** Nonlinear (quadratic)
3. $2x_1 - 7x_2 + 3x_3 = 15$ ✓Linear
4. $x_1 \cdot x_2 \geq 10$ **X** Nonlinear (bilinear)
5. $\sqrt{x_1} + x_2 \leq 8$ **X** Nonlinear (square root)

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6. $\frac{x_1}{5} + \frac{x_2}{3} \leq 1$

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5. $\sqrt{x_1} + x_2 \leq 8$ **X** Nonlinear (square root)
6. $\frac{x_1}{5} + \frac{x_2}{3} \leq 1$ ✓Linear (constants in denominator OK)

Classic LP Example: Product Mix

Scenario: A factory produces two products: A and B.

Data:

- Product A: \$40 profit/unit
- Product B: \$30 profit/unit
- Machine hours available: 100
- Labor hours available: 80

Resource Requirements:

	A	**B**
Machine	2 hrs	1 hr
Labor	1 hr	2 hrs

Question: How many of each product to make?

Product Mix - Mathematical Formulation

Decision Variables:

- x_1 = units of Product A
- x_2 = units of Product B

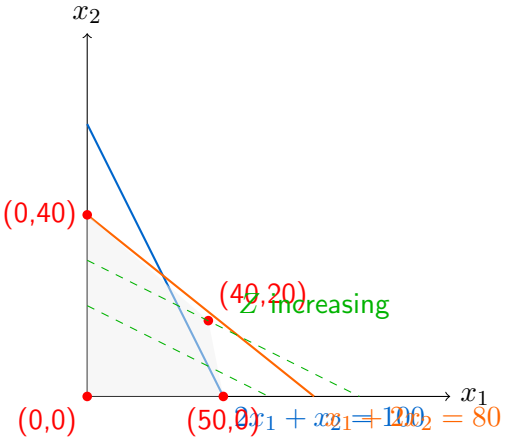
Objective Function:

$$\max Z = 40x_1 + 30x_2$$

Constraints:

$$2x_1 + x_2 \leq 100 \quad \text{(Machine hours)}$$
$$x_1 + 2x_2 \leq 80 \quad \text{(Labor hours)}$$
$$x_1, x_2 \geq 0 \quad \text{(Non-negativity)}$$

Geometric View of LP (2 Variables)



Key Insight: Optimal solution is always at a corner point (vertex) of the feasible region!

Key Properties of Linear Programming

1. **Feasible Region:** A convex polyhedron (or polygon in 2D)
2. **Optimal Solution:** Always at a vertex (corner point)
3. **Multiple Optima:** If two vertices are optimal, all points on the edge between them are optimal
4. **No Local Optima:** Any local optimum is a global optimum
5. **Efficient Algorithms:** Simplex method, Interior Point methods

These properties make LP very attractive for real applications!

Where is LP Used?

Manufacturing

- Product mix optimization
- Production planning
- Cutting stock problems
- Blending (oil refining, feed)

Logistics

- Transportation planning
- Distribution network design
- Warehouse location

Finance

- Portfolio optimization
- Risk management
- Asset allocation

Other Domains

- Diet planning
- Workforce scheduling
- Energy management
- Network flow

LP is the workhorse of optimization!

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When Linear Models Aren't Enough

Nonlinear Programming (NLP)

Optimization where the objective function or at least one constraint is **nonlinear**.

General Form:

$$
\begin{aligned}
\min \text{ or } \max \quad & f(x_1, x_2, \ldots, x_n) \\
\text{subject to} \quad & g_i(x_1, \ldots, x_n) \leq 0, \quad i = 1, \ldots, m \\
& h_j(x_1, \ldots, x_n) = 0, \quad j = 1, \ldots, p \\
& x \in \mathbb{R}^n
\end{aligned}
$$

where f, g_i, or h_j are nonlinear functions.

Common Nonlinear Functions

Polynomial

- Quadratic: $x^2 + 2xy + y^2$
- Cubic: $x^3 - 3x$
- Higher order

Exponential & Logarithmic

- e^{2x}, $\ln(x+1)$
- Growth/decay models

Trigonometric

- $\sin(x)$, $\cos(x)$
- Periodic phenomena

Rational

- $\frac{1}{x}$, $\frac{x^2}{y+1}$
- Rates and ratios

Nonlinear models can capture much richer behavior!

Why Do We Need Nonlinear Models?

Reality is Often Nonlinear!

1. **Economies of Scale**
   - Cost per unit decreases as volume increases
   - Cost $= 1000 + 5x - 0.01x^2$
2. **Diminishing Returns**
   - Adding more advertising has less and less impact
   - Sales $= 100 \ln(1 + \text{Ad Spend})$
3. **Physical Laws**
   - Energy: $E = \frac{1}{2}mv^2$ (quadratic in velocity)
   - Drag force: $F \propto v^2$
4. **Machine Learning**
   - Neural networks: nonlinear activation functions
   - Loss functions often nonconvex

NLP Example: Portfolio Optimization

Problem: Invest in stocks to maximize return while controlling risk.

Decision Variables:

- x_i = fraction of budget in stock i

Model:

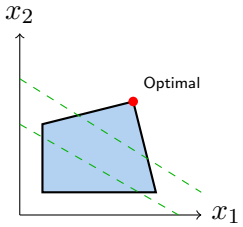
$$\begin{aligned} \max \quad & \sum_{i=1}^{n} r_i x_i - \lambda \sum_{i=1}^{n} \sum_{j=1}^{n} \sigma_{ij} x_i x_j \\ \text{s.t.} \quad & \sum_{i=1}^{n} x_i = 1 \quad \text{(invest all)} \\ & x_i \geq 0 \quad \text{(no shorting)} \end{aligned}$$

where:

- r_i = expected return of stock i
- σ_{ij} = covariance between stocks i and j
- λ = risk aversion parameter

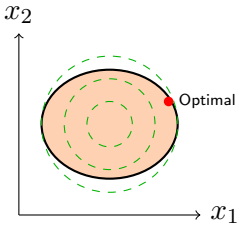
Geometric Difference: LP vs NLP

Linear Programming



Straight edges, corner optimal

Nonlinear Programming



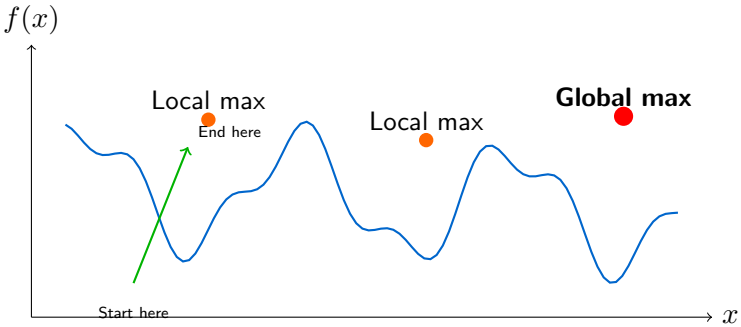
Curved boundaries, interior or boundary optimal

NLP: Optimal can be anywhere, not just corners!

Challenges in Nonlinear Programming

1. **Local vs Global Optima**
   - May get stuck at local optimum
   - No guarantee of finding global best
2. **Computational Complexity**
   - Much slower than LP
   - May not converge
3. **Need Good Starting Point**
   - Algorithms sensitive to initialization
   - Different starts may give different solutions
4. **No Universal Algorithm**
   - Different methods for different problem structures
   - Requires mathematical analysis

The Local Optimum Problem



Problem: Gradient-based methods climb to nearest peak, which may not be the highest!

Convex vs Nonconvex NLP

Convex NLP

- Objective and feasible region are convex
- Local optimum = Global optimum
- Efficiently solvable

Examples:

- $\min x^2 + y^2$
- Least squares regression
- Quadratic programming (QP)

Nonconvex NLP

- General nonlinear functions
- Multiple local optima possible
- Hard to solve globally

Examples:

- $\min \sin(x) + \cos(y)$
- Neural network training
- Engineering design

Convexity makes a huge difference in solvability!

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Side-by-Side Comparison

Property	**Linear (LP)**	**Nonlinear (NLP)**
Objective function	Linear	Nonlinear
Constraints	Linear	Can be nonlinear
Feasible region	Convex polyhedron	Any shape
Optimal location	Vertex (corner)	Anywhere
Local optima	None (always global)	Possible
Solution methods	Simplex, Interior Pt.	Gradient, Newton, etc.
Computational time	Fast (polynomial)	Can be slow
Guaranteed solution	Yes	Not always
Software	Many (CPLEX, Gurobi)	Many (IPOPT, SNOPT)

Decision Guide: LP or NLP?

Use LP when:

- Relationships are approximately linear
- Speed is critical
- You need guaranteed optimal solution
- Problem has many variables/constraints

Use NLP when:

- Nonlinearity is essential to the model
- Physical laws are inherently nonlinear
- Economies of scale or diminishing returns matter
- You can accept local optima or have good initialization

Hybrid Approach

Sometimes start with linear approximation, then refine with nonlinear model.

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Real World: Netflix Recommendations

Problem: Predict user ratings for movies.

Model: Matrix Factorization (Nonlinear!)

$$\min \sum_{(i,j)\in\text{observed}} (r_{ij} - \mathbf{u}_i^T \mathbf{v}_j)^2 + \lambda(|\mathbf{u}_i|^2 + |\mathbf{v}_j|^2)$$

where:

- r_{ij} = actual rating by user i for movie j
- $\mathbf{u}_i$ = user feature vector
- $\mathbf{v}_j$ = movie feature vector

Why Nonlinear?

- Quadratic terms in objective
- Bilinear term $\mathbf{u}_i^T \mathbf{v}_j$

This NLP powers 80% of what you watch on Netflix!

Real World: Tesla Battery Optimization

Problem: Optimize battery usage for maximum range.

Key Nonlinearities:

- Energy $= \frac{1}{2}mv^2$ (quadratic in speed)
- Drag force $\propto v^2$
- Battery efficiency varies with temperature
- Regenerative braking (nonlinear recovery)

Model Type:

- Nonlinear programming
- Constraints from physics (ODEs)
- Real-time optimization every few seconds

Impact

Optimal control increases range by 10-15% compared to simple strategies!

Real World: Google PageRank

Problem: Rank web pages by importance.

Model: Eigenvector problem (Nonlinear!)

$$\mathbf{r} = \alpha A\mathbf{r} + (1 - \alpha)\mathbf{e}$$

where:

- $\mathbf{r}$ = PageRank vector (what we solve for)
- A = link matrix
- α = damping factor (0.85)

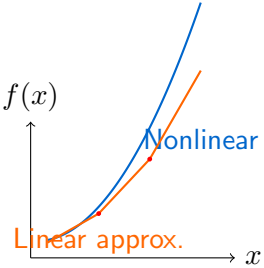
Why It's Optimization:

- Finding eigenvector = solving nonlinear system
- Can be cast as optimization problem
- Uses power iteration method

Billion-dollar algorithm built on nonlinear mathematics!

Linearizing Nonlinear Problems

Technique: Approximate nonlinear functions with piecewise linear.



Advantages:

- Can use fast LP solvers
- Guaranteed global optimum (of approximation)

Disadvantages:

- Only an approximation
- Need more variables (for breakpoints)

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Interactive Challenge: Classify These Problems

For each, identify: Linear or Nonlinear?

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   Linear
4. Maximizing revenue = price $\times$ quantity where price depends on quantity
   Nonlinear (bilinear if price $= f(\text{quantity})$)

Formulation Exercise: Fencing a Garden

Problem: You have 100 meters of fencing to enclose a rectangular garden. Maximize the area.

Your Task (3 minutes):

1. Define decision variables
2. Write objective function
3. Write constraints
4. Is this linear or nonlinear?

Work with your neighbor!

Fencing Solution

Decision Variables:

- x = length of garden
- y = width of garden

Objective:

$$\max \quad A = x \cdot y$$

Constraints:

$$2x + 2y = 100, \quad x, y \geq 0$$

Classification: Nonlinear! (objective is xy)

Solution: Using calculus or substitution, optimal is $x = y = 25$ (square), $A = 625$ m.

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Software Tools for LP

Commercial Solvers

- CPLEX (IBM)
- Gurobi
- XPRESS
- MOSEK

Free for academics!

Open Source

- GLPK (GNU)
- COIN-OR (CLP, CBC)
- HiGHS

Python Libraries

- PuLP
- CVXPY
- SciPy (linprog)

We'll use Python + CVXPY in this course.

Software Tools for NLP

General NLP Solvers

- IPOPT (open source)
- SNOPT
- KNITRO
- CONOPT

Convex Optimization

- CVX (MATLAB)
- CVXPY (Python)
- Mosek

Python Libraries

- SciPy (optimize)
- CVXPY
- PyOpt
- CasADi

Machine Learning

- TensorFlow
- PyTorch
- JAX

Choice depends on problem structure (convex vs nonconvex).

Python Example: Solving LP

```
import cvxpy as cp

# Decision variables
x1 = cp.Variable()
x2 = cp.Variable()

# Objective function
objective = cp.Maximize(40*x1 + 30*x2)

# Constraints
constraints = [
    2*x1 + x2 <= 100,     # Machine hours
    x1 + 2*x2 <= 80,      # Labor hours
    x1 >= 0,
    x2 >= 0
]

# Solve
problem = cp.Problem(objective, constraints)
problem.solve()

print(f"Optimal␣x1␣=␣{x1.value:.2f}")
print(f"Optimal␣x2␣=␣{x2.value:.2f}")
print(f"Maximum␣profit␣=␣${problem.value:.2f}")
```

Output: x1 = 40, x2 = 20, Profit = $2200

Python Example: Solving NLP

```
from scipy.optimize import minimize

# Objective: minimize x^2 + y^2
def objective(x):
    return x[0]**2 + x[1]**2

# Constraint: x + y = 10
def constraint(x):
    return x[0] + x[1] - 10

# Initial guess
x0 = [1, 1]

# Define constraint
con = {'type': 'eq', 'fun': constraint}

# Solve
solution = minimize(objective, x0, constraints=con)

print(f"Optimal x = {solution.x[0]:.2f}")
print(f"Optimal y = {solution.x[1]:.2f}")
print(f"Minimum value = {solution.fun:.2f}")
```

Output: x = 5, y = 5, Min = 50

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Today's Key Takeaways

1. **Linear models** have linear objectives and constraints
   - Easy to solve, guaranteed global optimum
   - Many real applications
2. **Nonlinear models** allow nonlinear functions
   - More realistic for many problems
   - Harder to solve, local optima possible
3. **Choice depends on problem**
   - Use simplest model that captures key features
   - Linear approximations often good enough
4. **Software makes both accessible**
   - Python + CVXPY/SciPy
   - Commercial solvers for large-scale

Questions?

Q & A Time